

Supplementary Material: Calibrate Once, Measure Within Support

SI Appendix

S1. Formal Definitions of Prevalence Estimation Methods

This section provides full mathematical definitions of the seven prevalence estimation methods compared in the simulation study.

Setup. Let $h(X) \in [0, 1]$ denote the device’s probabilistic prediction for input X , with true label $Y \in \{0, 1\}$. The goal is to estimate the target prevalence $\pi^* = P^*(Y = 1)$ using only unlabeled target data $\{X_i^*\}_{i=1}^n$ and calibration parameters estimated from a labeled source dataset.

S1.1 Uncalibrated Averaging

$$\hat{\pi}_{\text{raw}} = \frac{1}{n} \sum_{i=1}^n h(X_i^*)$$

S1.2 Classify & Count

Given a threshold τ chosen on calibration data:

$$\hat{\pi}_{\text{CC}} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[h(X_i^*) \geq \tau]$$

In the simulation, τ is chosen so that $\hat{\pi}_{\text{CC}}$ matches the true prevalence on the calibration set.

S1.3 Rogan-Gladen (Adjusted Count)

$$\hat{\pi}_{\text{RG}} = \frac{\hat{\pi}_{\text{CC}} - \widehat{\text{FPR}}}{\widehat{\text{TPR}} - \widehat{\text{FPR}}}$$

where $\widehat{\text{TPR}}$ and $\widehat{\text{FPR}}$ are estimated from calibration data at threshold τ (Rogan and Gladen 1978).

S1.4 Probabilistic Adjusted Classify & Count (PACC)

$$\hat{\pi}_{\text{PACC}} = \frac{\bar{h} - \hat{\mu}_0}{\hat{\mu}_1 - \hat{\mu}_0}$$

where $\bar{h} = \frac{1}{n} \sum_i h(X_i^*)$, $\hat{\mu}_1 = \hat{\mathbb{E}}[h(X)|Y = 1]$, and $\hat{\mu}_0 = \hat{\mathbb{E}}[h(X)|Y = 0]$ are estimated from calibration data (González et al. 2017).

S1.5 SLD (EMQ)

The Sacerens-Latinne-Decaestecker algorithm iterates:

1. Initialize $\hat{\pi}^{(0)}$ from the source prevalence.
2. E-step: Adjust posteriors for the new prior:

$$\tilde{h}_i^{(t)} = \frac{(\hat{\pi}^{(t)}/\pi_s) \cdot h(X_i^*)}{(\hat{\pi}^{(t)}/\pi_s) \cdot h(X_i^*) + ((1 - \hat{\pi}^{(t)})/(1 - \pi_s)) \cdot (1 - h(X_i^*))}$$

3. M-step: $\hat{\pi}^{(t+1)} = \frac{1}{n} \sum_i \tilde{h}_i^{(t)}$
4. Repeat until convergence (Sacerens et al. 2002).

S1.6 Global Calibration

In the simulation, global calibration applies a multiplicative correction:

$$h_{\text{cal}}(X) = c \cdot h(X)$$

where $c = \bar{Y}_{\text{cal}}/\bar{h}_{\text{cal}}$ is estimated on calibration data. This is the curve labeled “global recalibration” in Figure 1 and Figures S1–S2; it is multiplicative rescaling, not isotonic regression. (In the simulation the two would coincide because the score takes only two values, each mapped to its stratum frequency; we therefore report the multiplicative version and reserve the isotonic-vs-feature-conditional contrast for the empirical applications, where scores coarsen the features.) In the empirical applications, global calibration uses isotonic regression on the probability scores.

S1.7 Multicalibration

In the simulation, which has a single binary covariate, multicalibration reduces to stratum-specific additive corrections:

$$h_{\text{mc}}(X) = h(X) + \hat{\epsilon}_g \quad \text{for } X \in \text{stratum } g$$

where $\hat{\epsilon}_g = \bar{Y}_g - \bar{h}_g$ is estimated on calibration data within each stratum.

In the empirical applications, we use MCGrad (Tax et al. 2026), a multicalibration algorithm based on gradient boosting. MCGrad operates in logit space: given a base predictor $f_0(X)$ with logit $F_0(X) = \text{logit}(f_0(X))$, it iteratively fits gradient boosted decision trees (GBDTs) on the residuals between labels and current predictions. At each round t , a GBDT g_t is trained with the current logit predictions as `init_score` and with the feature matrix consisting of the segment features (categorical and numerical) augmented by the current logit prediction as an additional input feature. The logit predictor is then updated as $F_{t+1}(X) = \alpha_t \cdot (F_t(X) + g_t(X))$, where α_t is a single scalar unshrinkage factor estimated by a one-parameter logistic regression of the labels on the combined logit $F_t(X) + g_t(X)$, applied to the full running logit rather than to the increment alone. Because α_t is fit to maximize fit of the combined logit to the labels at each round, it counteracts the shrinkage induced by the GBDT learning rate without disturbing the relative structure the trees discovered; Tax et al. (2026) establish that the resulting sequence converges to a multicalibrated predictor. By including the prediction as a feature, GBDT splits naturally discover miscalibrated regions in the joint space of features and score levels, thereby approximating multicalibration without requiring

explicit group specification. Early stopping on a validation set prevents overfitting. MCGrad uses LightGBM as the GBDT implementation. See Tax et al. (2026) for convergence results and deployment details. Our claims are algorithm-agnostic: any procedure producing a multi-accurate, a fortiori multicalibrated, predictor over the feature class delivers the same guarantee, and several exist (Hébert-Johnson et al. 2018; Gopalan et al. 2022; Detommaso et al. 2024). An open-source implementation of MCGrad is available at <https://mcgrad.dev>.

S1.8 Multi-accuracy versus multicalibration

The main text states the universal-adaptability guarantee for a multicalibrated predictor and computes one with MCGrad. The property strictly required, however, is weaker. A predictor f is *multi-accurate* with respect to a group class $\mathcal{G}$ if $\mathbb{E}[f(X) - Y \mid X \in G] = 0$ for every $G \in \mathcal{G}$: its signed error averages to zero within each group, though it need not be calibrated at each score level within a group. Multi-accuracy alone is sufficient for unbiased prevalence on every target reweighting captured by $\mathcal{G}$: the universal-adaptability identity in the main text goes through verbatim, because it uses only that the within-cell error ϵ_G vanishes, which multi-accuracy guarantees (Kim et al. 2022). (For a single fixed target, only the still weaker $\sum_G w_G^* \epsilon_G = 0$ is needed; multi-accuracy across all of $\mathcal{G}$ is what makes one instrument valid for *every* target, i.e., target-independent.) Multi-accuracy does not by itself imply pointwise accuracy $\mathbb{E}[f(X) \mid X = x] = \mathbb{E}[Y \mid X = x]$ unless $\mathcal{G}$ is rich enough to separate the relevant regions of X (in the limit, singletons).

Multi-accuracy is therefore the minimal sufficient condition, but multicalibration is the better practical target, for three reasons. First, practical post-hoc algorithms produce multicalibrated predictors at no extra cost: MCGrad (Tax et al. 2026), which we use, fits the residual structure of X by gradient boosting and naturally approximates multicalibration (Section S1.7). Second, multicalibration extends the guarantee to cases where multi-accuracy is insufficient: when the shift is mediated by the device’s own scores (e.g., score-based selection of documents to hand-code), when prevalence is estimated within score strata, or when scores enter a downstream regression. Third, a predictor multicalibrated over a rich group class $\mathcal{G}$ is automatically multi-accurate over any subgroup measurable with respect to $\mathcal{G}$, providing robustness to misspecifying which of the calibrated features drive the shift.

S1.9 Estimation and implementation details

Simulation. Each of the 50 runs uses $n = 10,000$ observations, with fresh calibration data drawn at $P(X = 0) = 0.5$; bias and RMSE are evaluated on 20 target distributions with $P(X = 0) \in [0.01, 0.99]$. Definitions of all seven methods are in Section S1.

Comparative Agendas Project. The two campaigns (binary Yes/No and direct probability elicitation) were run separately to avoid anchoring. Benchmark implementations: Rogan-Gladen uses TPR/FPR estimated on the calibration set; IPW estimates density ratios by logistic regression on country, document type, decade, and length; isotonic regression is fit on the scores. Per-scenario bias is in Table S2; the ReadMe comparison is in Section S7 and the Llama 3.3 70B replication in Section S2.

American Community Survey. Training states are TX, MI, PA, OH, IL, GA, NC, VA (2016–2018); held-out test states are CA, NY, FL, WA, AZ, CO, with in-distribution test $n \approx 920,000$. Age-shifted targets are produced by importance-weighted resampling, and RMSE by a 200-iteration

bootstrap. Post-hoc calibration uses isotonic regression and MCGrad with categorical and numerical features.

S2. Robustness: Replication with Open-Weight LLM (Llama 3.3 70B)

The main text reports results using Claude Opus 4.6 as the LLM measurement device. To verify that the findings are not specific to a particular model, we replicate the CAP analysis using Llama 3.3 70B Instruct (Grattafiori et al. 2024) (4-bit NF4 quantized, run on a single A100 80GB GPU). This section reports results using two score extraction methods: token log-probabilities and verbalized confidence elicitation.

S2.1 Score Extraction Methods

Log-probabilities. For each document, the model is prompted with the CAP codebook definition of Law & Crime and asked to respond Yes or No. The score is extracted from next-token log-probabilities: $h(X) = P(\text{Yes}) / (P(\text{Yes}) + P(\text{No}))$. This produces highly bimodal scores: 23% of the 105,000 documents score at exactly 0.0 or 1.0, and only 6% fall in the mid-range [0.1, 0.9]. Because MCGrad’s internal logit transform maps values near 0 and 1 to $\pm\infty$, a linear squashing transformation $h'(X) = \epsilon + (1 - 2\epsilon) \cdot h(X)$ with $\epsilon = 0.05$ is applied before fitting MCGrad.

Verbalized confidence (2-stage). A two-stage dialogue first asks the model to classify the document (Yes/No), then asks it to estimate the probability that its answer is correct, with an anti-certainty instruction (“Note: very few things are 0% or 100% certain”) to discourage degenerate outputs (Tian et al. 2023). The score is $P(\text{correct})$ if the answer is Yes and $1 - P(\text{correct})$ if No. This produces scores in [0.01, 0.99] with negligible boundary mass and 11 unique score values. No squashing is required.

S2.2 Data and Calibration

The Llama analysis uses the full 105,000-document sample (15,000 per sub-population for Denmark questions, Spain questions, U.S. bills, and Belgium newspaper; 30,000 for Spanish media; 15,000 for Belgian TV). The calibration set ($n \approx 40,000$) is drawn equally from the four in-distribution sub-populations. MCGrad is calibrated with categorical features (country, document type, party) and one numerical feature (decade). Because this analysis draws on the larger 105,000-document sample rather than the 30,000-document Opus main-text sample, the expert-coded true prevalences below differ slightly from Table S2 (e.g., 8.1% vs. 7.9% at baseline); these are sampling differences in the gold standard, not discrepancies in the method.

S2.3 Results: Verbalized Confidence Scores

Table S3 shows prevalence estimation bias using Llama 3.3 70B with verbalized confidence scores.

Scenario	Shift Type	True Prev.	CC	RG	IPW	Iso.	MCGrad
Baseline	None	8.1%	+14.7	+0.6	+0.1	+0.2	+0.2

Scenario	Shift Type	True Prev.	CC	RG	IPW	Iso.	MCGrad
Country shift	Within-cal.	8.7%	+15.6	+2.2	-0.3	+1.3	+0.1
Doc-type shift	Within-cal.	6.5%	+16.0	+1.7	+0.0	+0.7	+0.1
Spain media	OOD doc type	19.3%	+15.4	+6.6	-9.8	-7.2	-4.9
Belgium TV	OOD doc type	11.1%	+13.3	-0.0	-3.5	-1.6	-3.4

Table S3: Prevalence estimation bias (pp) for Law & Crime topic using Llama 3.3 70B with verbalized confidence scores. CC = Classify & Count, RG = Rogan-Gladen, IPW = importance-weighted estimation, Iso. = isotonic regression.

The pattern is consistent with the main text’s Claude Opus results: MCGrad achieves near-zero bias within the calibration distribution (≤ 0.2 pp) and degrades on OOD populations (-3.4 to -4.9pp). Several differences are notable:

- **Higher raw CC bias** (+14-16pp vs. +2-5pp with Opus).
- **Comparable MCGrad within-calibration performance** (≤ 0.2 pp for both models), confirming that multicalibration corrects for model-specific calibration errors.
- **Larger OOD bias** on Spanish media (-4.9pp vs. Opus’s -1.9pp on binary labels / -4.5pp on probability scores), reflecting the combination of a weaker base model with a coarser score distribution. Note that this comparison mixes elicitation modes: the Llama numbers here use verbalized confidence (11 unique values), whereas the Opus main-text headline uses binary labels; the closest like-for-like comparison is Llama verbalized against Opus’s probability-score condition (43 unique values, SI Figure S3), and on that comparison the gap is consistent with the finer Opus score distribution. The broader point (that MCGrad’s within-calibration performance is near-identical across models and elicitation modes while the input score distribution matters chiefly out of support) holds in every comparison.

S2.4 Score Distribution: Log-Probabilities vs. Verbalized Confidence

The bimodal distribution of Llama’s log-probability scores illustrates a broader challenge for LLM-based measurement. RLHF-tuned instruction-following models tend to produce highly confident outputs, pushing token probabilities toward 0 or 1. This creates two problems for prevalence estimation: (1) the scores carry little information about uncertainty, producing large raw bias even at baseline (+18pp), and (2) post-hoc calibration methods that operate in logit space (including MCGrad) require score preprocessing to avoid numerical instability.

Verbalized confidence elicitation partially addresses both problems by producing scores that are better distributed (75% in $[0.1, 0.9]$) and better calibrated out of the box (log loss 0.525 vs. 1.707 for log-probabilities). However, the scores remain coarsely discretized (11 unique values), and as shown in both the Llama and Opus analyses, the quality of the input scores matters less than the metadata features for MCGrad’s prevalence estimation performance under shift.

S2.5 Additional Baselines: SLD and PACC on Llama Scores

The SLD (EMQ) algorithm, designed for label shift rather than covariate shift, diverges catastrophically on Llama’s verbalized confidence scores, producing prevalence estimates biased by +33 to +60pp. This occurs because the verbalized scores are not calibrated posteriors, violating SLD’s core assumption. PACC shows moderate bias (+0.6 to +5.9pp within calibration, +2.4 to +5.9pp OOD). Full results including SLD and PACC are available in the replication code.

S3. Detailed Results Tables

Table S1: ACS Employment Prevalence Estimation Bias

Setting	Age Dist.	True Prev.	Raw	CC	RG	PACC	SLD	IPW	Iso.	MCGrad
In-Dist	Original	46.0%	-0.31	-0.07	+0.26	-0.07	+0.01	-0.3	-0.30	-0.27
In-Dist	Young-skewed	12.8%	+1.93	+2.47	-12.82	-12.82	-11.95	-0.3	+2.04	-0.11
In-Dist	Old-skewed	16.8%	+7.23	-6.62	-16.77	-16.77	-16.76	-1.2	+6.65	+0.22
In-Dist	Bimodal	21.1%	+4.57	-1.50	-18.62	-19.97	-16.14	+4.7	+4.33	+0.12
OOD	Original	45.1%	+1.15	+1.40	+2.12	+2.13	+2.25	+1.7	+1.17	+1.35
OOD	Young-skewed	13.0%	+2.93	+3.73	-12.96	-12.96	-11.38	+0.8	+3.08	+0.88
OOD	Old-skewed	16.0%	+8.47	-5.64	-15.97	-15.97	-15.97	+0.1	+7.91	+1.01
OOD	Bimodal	20.8%	+5.91	+0.09	-16.14	-17.27	-15.20	+6.3	+5.69	+1.13

Prevalence estimation bias in percentage points (pp) under synthetic age distribution shift. Raw = uncalibrated averaging, CC = Classify & Count, RG = Rogan-Gladen, IPW = importance-weighted prevalence estimation, Iso. = Isotonic regression. Bootstrap RMSE (200 iterations) closely tracks absolute bias in all scenarios.

Table S2: CAP Law & Crime Prevalence Estimation Bias (Claude Opus 4.6)

Scenario	Shift Type	True Prev.	CC	RG	SLD	IPW	Iso.	MC (binary)	MC (scores)
Baseline	None	7.9%	+2.2	+0.5	+7.4	+0.1	+0.1	+0.1	+0.2
Country shift	Within-cal.	8.4%	+3.3	+1.7	+9.5	+0.1	+0.9	+0.4	+0.4
Doc-type shift	Within-cal.	6.3%	+1.6	-0.3	+4.9	+0.1	+0.0	-0.0	+0.1

Scenario	Shift Type	True Prev.	CC	RG	SLD	IPW	Iso.	MC (binary)	MC (scores)
Spain media	OOD doc type	19.5%	+3.6	+3.1	+21.6	-12.2	-2.5	-1.9	-4.5
Belgium TV	OOD doc type	11.1%	+4.8	+3.7	+13.7	-4.5	+2.6	+0.7	+1.6

CC = Classify & Count (fraction of Yes labels); RG = Rogan-Gladen adjustment on binary labels; SLD = Saerens-Latinne-Decaestecker (label shift, applied to probability scores); IPW = importance-weighted estimation (target-specific density ratio); Iso. = isotonic regression on probability scores; MC (binary) = MCGrad on binary labels with base-rate initialization; MC (scores) = MCGrad on probability scores.

S4. Simulation: RMSE

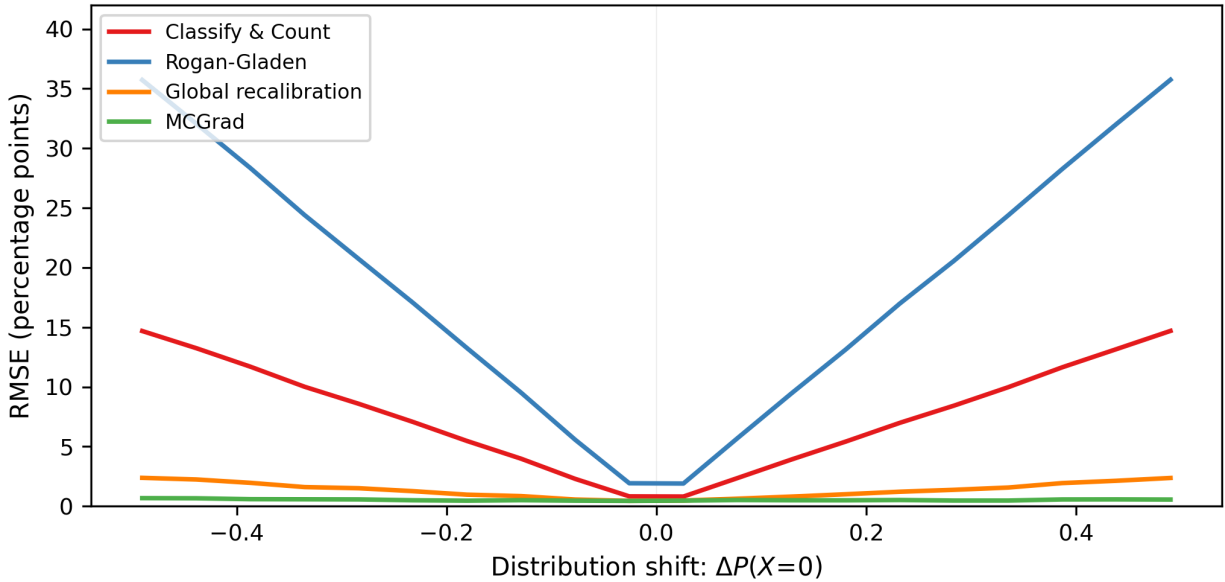


Figure S1: Root mean squared error (RMSE) under covariate shift for the same four methods shown in Figure 1 (Classify & Count, Rogan-Gladen, multiplicative global recalibration, MCGrad), averaged over 50 simulation runs. RMSE closely tracks absolute bias for all methods, confirming that variance is small relative to bias at this sample size. MCGrad maintains the lowest RMSE across all shift levels.

S5. Simulation: All Methods

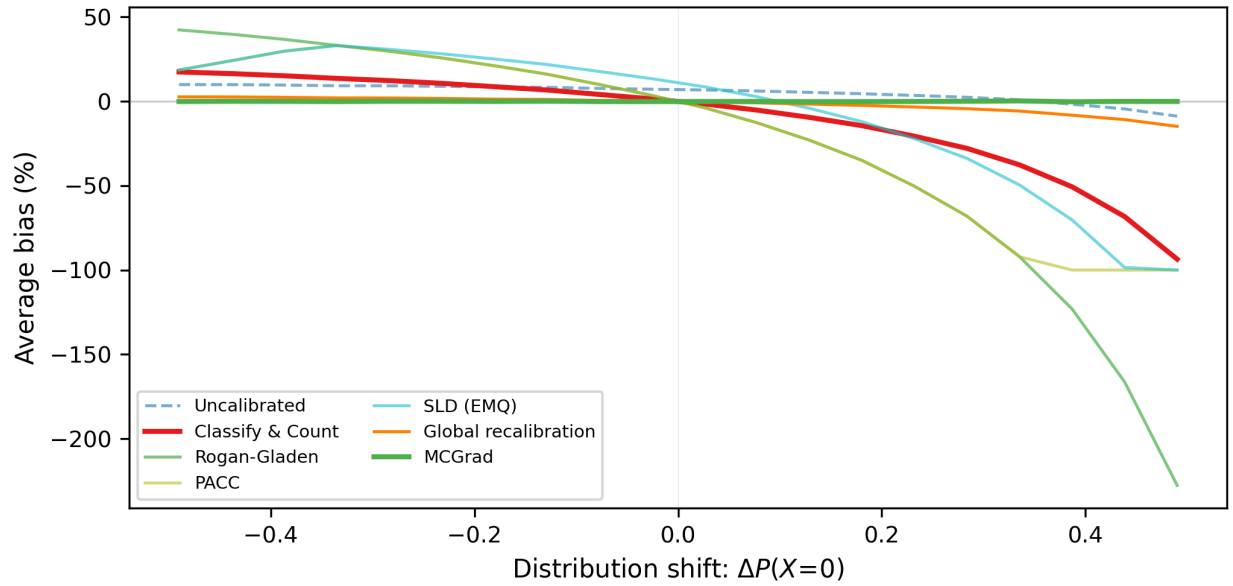


Figure S2: Simulation bias curves for all seven methods (uncalibrated averaging, Classify & Count, Rogan-Gladen, PACC, SLD/EMQ, multiplicative global recalibration, MCGrad). Rogan-Gladen and PACC exhibit catastrophic failure (bias exceeding -200% at extreme shifts). SLD shows large bias under covariate shift because it assumes label shift. Multiplicative global recalibration shows moderate bias, up to roughly 15% at the most extreme shift. MCGrad maintains near-zero bias throughout.

S6. Claude Opus Score Distribution

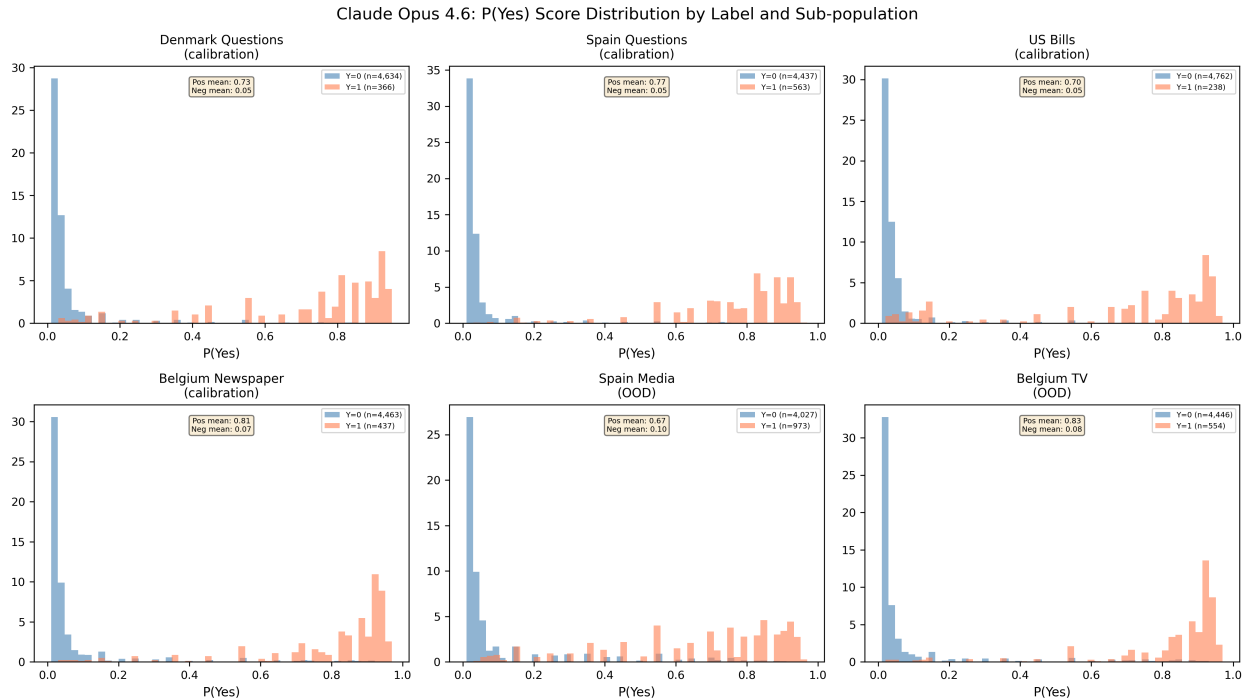


Figure S3: Claude Opus 4.6 $P(\text{Yes})$ score distribution by label across six CAP sub-populations. Scores are well-separated (mean 0.75 for positives vs. 0.07 for negatives) with 43 unique values and no boundary mass.

For reference, the discrimination figures reported in the main text are tabulated here. Pooled AUC is 0.960 for the binary (Yes/No) condition and 0.987 for the probability-score condition; per-language AUCs range from 0.983 to 0.994. The probability-score condition has 43 unique values (this figure); the verbalized-confidence elicitation used in the Llama replication produces 11 unique values (SI Section S2). All discrimination metrics are computed against the CAP expert codes as the gold standard.

S7. Relationship to the ReadMe Family of Quantifiers

The political-science quantification literature is anchored by ReadMe (Hopkins and King 2010) and its successor ReadMe2 (Jerzak et al. 2023), which estimate category proportions directly from document-feature distributions without per-document classification. These methods, like the classical Saerens-Latinne-Decaestecker (SLD) algorithm (Saerens et al. 2002) we report in Tables S1–S2 and the distribution-matching quantifiers surveyed by González et al. (2017), assume *label shift*: the class-conditional feature distribution $P(X | Y)$ is stable across the labeled and target sets while the class prior $P(Y)$ may change. ReadMe2 improves robustness to differences in the document-feature distribution between labeled and target sets, but does so within this label-shift framing and is estimated for a given labeled/target pair.

The regime studied in the main text is the opposite: *covariate shift*, where $P(X)$ changes across populations while $P(Y | X)$ is stable, the natural model when document features (language, venue, era, content) drive the category rather than the reverse. Under covariate shift the within-class feature distribution $P(X | Y)$ is no longer stable, so the label-shift identity these methods solve is misspecified.

We test this directly by benchmarking ReadMe on the CAP corpora, using the same calibration set and shift scenarios as Table S2. We report two implementations on the identical documents: a from-scratch implementation of the Hopkins-King estimator (binary word-presence features summarized over 300 random feature subsets of size 15; `cap_analysis/readme_baseline.py`), and the official IQSS `readme` package’s classic estimator (`readme0`; 200 subsets, vocabulary 2,500). Both are classifier-free: they use the document text and the calibration labels, not the LLM’s outputs.

Scenario	Shift	True	ReadMe (ours)	ReadMe (official <code>readme0</code>)
Baseline	none	7.9%	+0.8	+5.3
Country shift	within-cal.	8.4%	+67.6	+60.1
Doc-type shift	within-cal.	6.3%	+7.1	-5.0
Spain media	OOD doc type	19.5%	+4.5	+44.5
Belgium TV	OOD doc type	11.1%	+13.5	+51.5

Table S4: ReadMe prevalence-estimation bias (pp) on the CAP corpora; identical calibration set and scenarios as Table S2. Both estimators are classifier-free (text only). For comparison, MCGrad (Table S2) stays within 1.9 pp across all scenarios.

The two implementations agree on direction and differ only in magnitude (which depends on vocabulary, feature weighting, and profile-matching choices). ReadMe is near-unbiased *only* without shift (our implementation attains +0.8pp at baseline, validating it, while the official `readme0` is already +5 to +7pp at baseline on this short, multilingual corpus) and is biased and highly unstable under every shift, severely so out of support (the official estimator reaches +44 to +60pp on the held-out genres; the within-calibration country shift exceeds +60pp in both implementations). The instability is intrinsic: across random feature subsets our implementation’s per-estimate standard deviation is 11pp at baseline and 32–42pp under shift. The mechanism is interpretable: ReadMe’s binary word-presence profiles are sensitive to document length, so when the target’s mean feature density departs from the calibration set’s (long Belgian newspaper documents activate roughly twice as many features as the calibration average; short Spanish-media and Belgian-TV snippets far fewer), the within-class feature distribution shifts and the label-shift identity breaks. This is the covariate-shift failure the theory predicts, shown directly for the canonical political-science quantifier rather than inferred from SLD.

ReadMe2 (Jerzak et al. 2023) refines the feature summary (word-vector “vector summaries” and a learned matched projection) but estimates proportions under the same label-shift assumption, so in principle it cannot escape this failure. We installed and ran the official ReadMe2 estimator; a faithful evaluation requires its `undergrad` vector-summary features, and when supplied instead with multilingual-BERT document embeddings the estimator was biased by roughly +20pp even at the no-shift baseline, mis-specified for that feature representation and the rare (~8%) positive class, so we do not report it as a fair head-to-head comparison. The defensible empirical statement is the classic-ReadMe result in Table S4; ReadMe2’s refinements target feature quality, not the invariance that fails under covariate shift.

S8. Disclosure of Generative-AI Use

The authors disclose the use of generative AI in the research and writing process. Under the GAIDeT taxonomy (2025), the following tasks were delegated to GAI tools under full human supervision: literature search and systematization, code generation and optimization, data collection and cleaning, data analysis, visualization, reproducibility testing, text generation, proofreading and editing, reformatting, and identification of limitations. The GAI tools used were Claude Opus 4.6, Claude Opus 4.7, and Gemini 3 Pro. Note that Claude Opus 4.6 is also the measurement device under study in the main CAP application; its role there is as an object of analysis, not as a research assistant. No task was performed exclusively by AI; all outputs were verified and iterated on by the authors, who bear sole responsibility for the manuscript. GAI tools are not authors. Declaration submitted by: Fridolin Linder.

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